

Differentials

Date _____ Period _____

For each problem, find the differential dy .

1) $y = -x^3 - 2$

2) $y = -\frac{3}{x}$

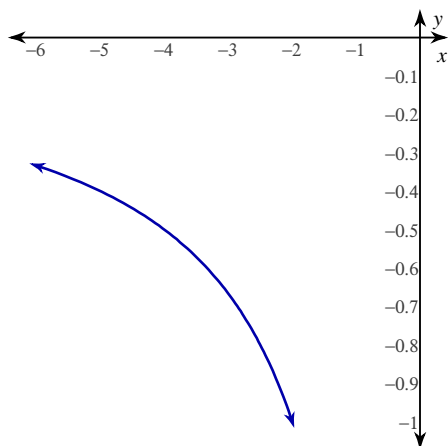
For each problem, find the general formulas for dy and Δy .

3) $y = -x^3 - 2$

4) $y = \frac{2}{x}$

For each problem, find dy and Δy , given x_0 and $dx = \Delta x$. You may use the provided graph of the function to sketch dx , Δx , dy , and Δy .

5) $y = \frac{2}{x}$; $x_0 = -5$, $dx = \Delta x = \frac{5}{2}$



6) $y = -x^3 + 1$; $x_0 = -1$, $dx = \Delta x = -\frac{1}{2}$

